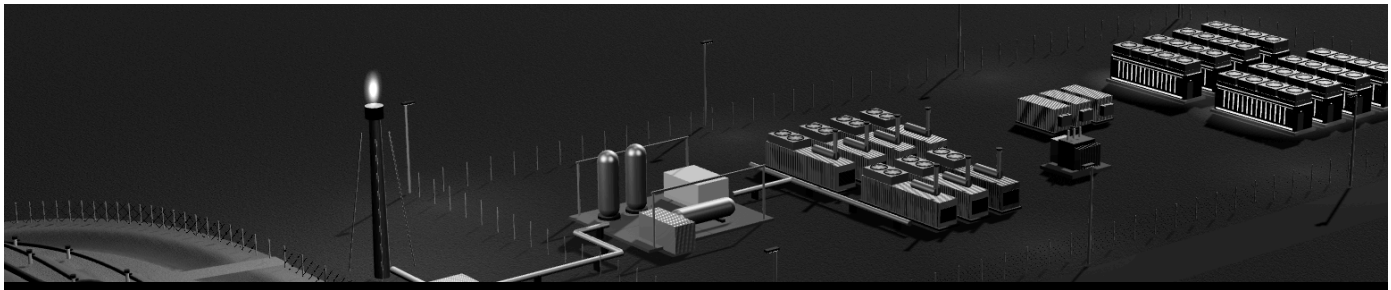
Atlantic Waste Disposal landfill, Waverly, Virginia: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

8.7 MW	8	15 ac	801k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $600k	$15k to $31k	$15k to $631k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=6093&CH4=26.1

THE MATH

Landfill gas flared beside the leachate evaporator (2022)	6,093 scfm
Methane content (2023)	26.1%
Energy in the gas	96.2 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 8.7 MW
Compute (power less cooling and losses, /1.2)	about 7.3 MW
Compute modules (about 1 MW each)	8
Gas engines (running plus one spare)	7 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 15 acres
Gas bought per year (95% run time)	801k MMBtu
Gas payment, up to $0.75/MMBtu	up to $600k
Land lease at $1,000 to $2,000 per acre per year	$15k to $31k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Lean-gas engines for 26% methane, and any existing gas agreement
- Virginia DEQ air permit (9VAC5-80)
- Tie-in under the landfill's Title V permit (NSPS Subpart XXX, NESHAP Subpart AAAA)
- County or municipal zoning for the data hall

Source: EPA Greenhouse Gas Reporting Program, 2023: about 8,090 scfm collected at 26% methane. EPA Landfill Methane Outreach Program: about 6,129 scfm flared in 2022 beside the leachate evaporator. Lean gas: lean-gas engines needed. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.